UNIVERSITI TEKNOLOGI MALAYSIA, JOHOR BAHRU

FACULTY OF COMPUTING
SEMESTER 2, SESSION 2025/2026

SECJH - BACHELOR OF COMPUTER SCIENCE

GROUP ASSIGNMENT
SMARTFOUND: SMART SOLUTION FOR LOST AND FOUND IN UTM

SECJ3483-03 TEKNOLOGI WEB (WEB TECHNOLOGY)

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1.0 Problem Statement and User Focus

1.1 Background

Universiti Teknologi Malaysia (UTM) Johor Bahru is one of Malaysia's largest and most active university campuses. As of early 2026, UTM hosts over 20,000 students on its main Johor Bahru campus alone, supported by approximately 4,821 staff members comprising academic and support personnel. With such a large and diverse community, lost and found incidents occur regularly across campus facilities including lecture halls, libraries, cafeterias, laboratories, and residential colleges. Currently, UTM has no official or dedicated platform to handle lost and found cases digitally. Students and staff rely entirely on informal channels such as WhatsApp groups, Telegram channels, and social media posts to report or search for missing items.

1.2 Problem Statement

The absence of a centralized lost and found system at UTM Johor Bahru creates several key problems:

1. Scattered information : lost and found posts are spread across multiple unofficial platforms with no single place to search
2. No verification process : anyone can claim an item with no structured way to verify legitimate ownership
3. No moderation : there is no designated authority to oversee or manage the process
4. Posts get buried : items posted on WhatsApp or social media quickly disappear as new messages arrive
5. Low recovery rate : without a proper system, lost items rarely reach their rightful owners

SmartFound is proposed as a dedicated, role-based web platform that centralizes the entire lost and found process for the UTM community. It provides a structured workflow from posting an item, submitting a claim, verifying ownership, and resolving the case all managed within one accessible platform.

1.3 Target Users

User Role	Who They Are	What They Can Do
User	UTM students and staff	Browse all posts, register and login, post lost or found items, submit claims, manage own posts
Officer	Designated UTM staff	Everything a User can do, plus verify and approve or reject claims, update item status
Admin	System administrator	Full system access, manage users and officers, delete any post, monitor all activity, view dashboard statistics

2.0 Functional Scope and Modules

SmartFound is divided into five main modules, where each module is designed to solve specific issues in the current UTM lost and found process. The system supports three types of users, which are Users (UTM students and staff), Officers (assigned UTM staff), and Admins. These modules help manage the entire lost and found process, starting from reporting an item until the case is successfully resolved.

Module	Features	User Actions	Role Access
M1 – Authentication & User Management			
Registration & Login	Register with username and passwordLogin with session-based authenticationLogout from any page	Submit registration formLogin to access protected pagesLogout via navbar	User, Officer, Admin
Profile Management	View own profile detailsUpdate full nameChange password	Edit profile on My Profile page	User, Officer, Admin
Role Assignment	Roles: Student (1), Admin (2)New registrations default to Student roleAdmin can delete user accounts	Admin views and manages all users via All Users pageAdmin deletes user accounts	Admin only
M2 – Lost & Found Item Reporting			
Submit Report	Post a Lost or Found reportFill item name, description, date, category, location, contact numberUpload an item imageReport auto-assigned OPEN status on creation	Complete and submit Add Report form	User, Officer, Admin
Categorisation & Location	10 item categories (Phone, Wallet, Keys, Student ID, Bag, Laptop, Book, Clothing, Water Bottle, Electronics)19 UTM	Select category and location from dropdown during report submission	User, Officer, Admin

	campus locations (faculties, libraries, parks/cafeterias)		
Manage Own Reports	View all personal reportsUpdate status from OPEN to CLOSED (item resolved)View full report detail including image	Access My Reports pageClick Update Status button to close a report	User, Officer, Admin
M3 – Browse & Discovery			
All Reports Feed	Browse all OPEN lost and found postsFilter by category and locationView item images, descriptions, dates, and contact infoSorted by newest first	Scroll the All Reports pageApply category/location filters	User, Officer, Admin
Report Detail View	View full report details on a dedicated pageSee all associated commentsView report status badge (OPEN / CLOSED)	Click on a report card to open report.php detail view	User, Officer, Admin
Closed Reports Archive	View all resolved (CLOSED) reportsProvides historical record of recovered items	Browse Closed Reports page	User, Officer, Admin
M4 – Communication & Claiming			
Comment System	Leave a comment on any OPEN reportView all comments with username and timestampComments disabled once report is CLOSED	Submit comment via form on Report detail pageComment stored with user_id and timestamp	User, Officer, Admin
Contact Information	Reporter's contact number visible on each report detailEnables	View contact number on report detail page	User, Officer, Admin

	direct communication between finder and owner		
M5 – Administration & Moderation			
Report Management	View all reports across the platformDelete any report (e.g. spam, violations)Access item images from any report	Access Manage Reports page (Admin-only route)Click Delete on any report	Admin only
User Management	View all registered users with role badgesDelete any user accountRole display: Student or Admin	Access All Users page (Admin-only route)Click Delete on any user	Admin only
Access Control	Admin-only pages protected by role check (role_id = 2)Session-based authentication throughoutNon-authenticated users redirected to login	Enforced server-side on every restricted page	Admin only

2.1 Role-Feature Access Matrix

The table below summarises which features are accessible by each role in SmartFound.

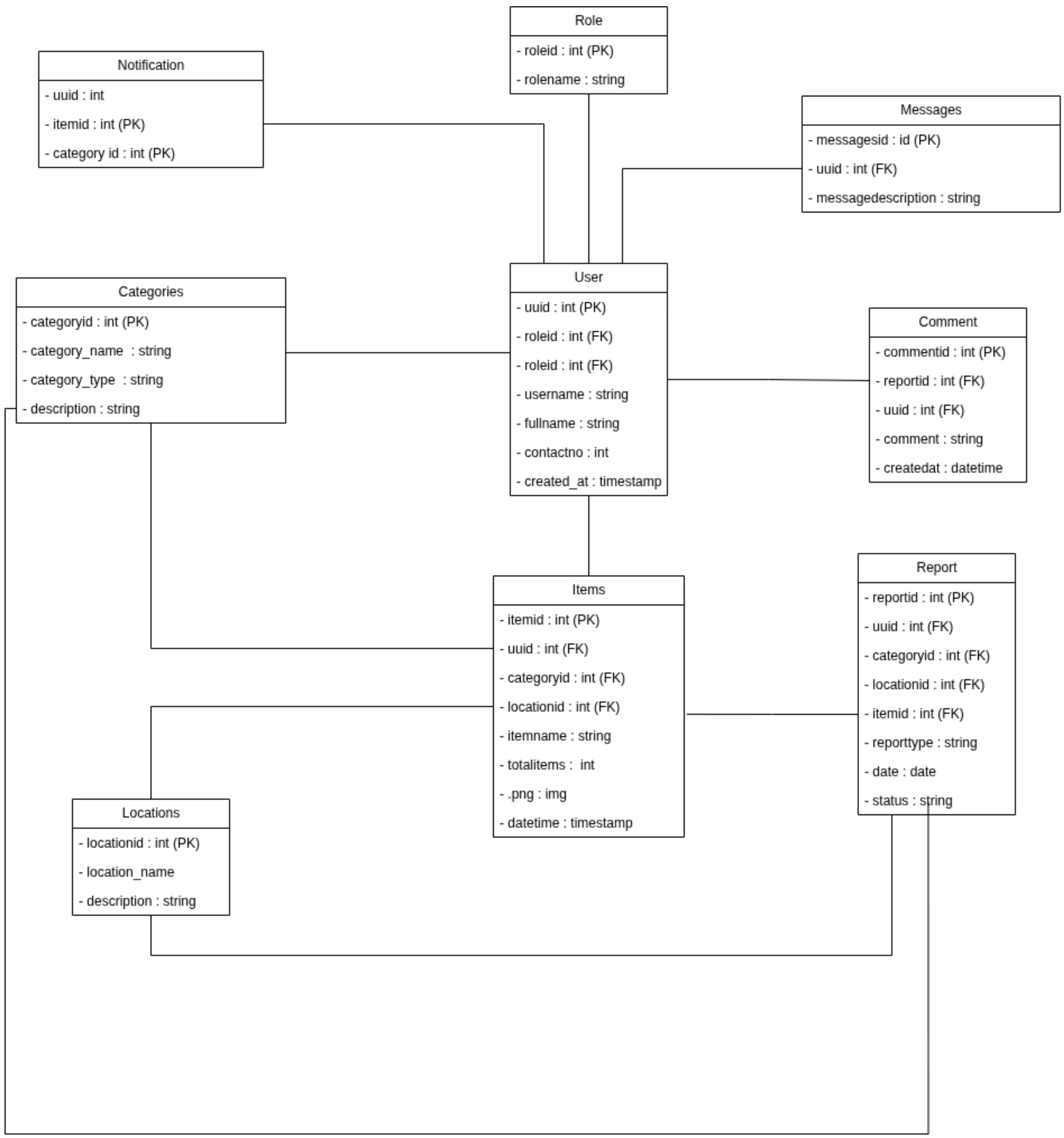
Feature	User	Officer	Admin
Register and Login	✓	✓	✓
Update own profile and password	✓	✓	✓
Browse all OPEN reports	✓	✓	✓
View report detail and comments	✓	✓	✓
Submit Lost or Found report	✓	✓	✓
Upload item image	✓	✓	✓
Manage and close own reports	✓	✓	✓
Comment on OPEN reports	✓	✓	✓
View closed reports archive	✓	✓	✓
Delete any report	✗	✓	✓
View and manage all users	✗	✗	✓
Delete any user account	✗	✗	✓
Access admin-protected pages	✗	✗	✓

2.2 Supporting Databases Table

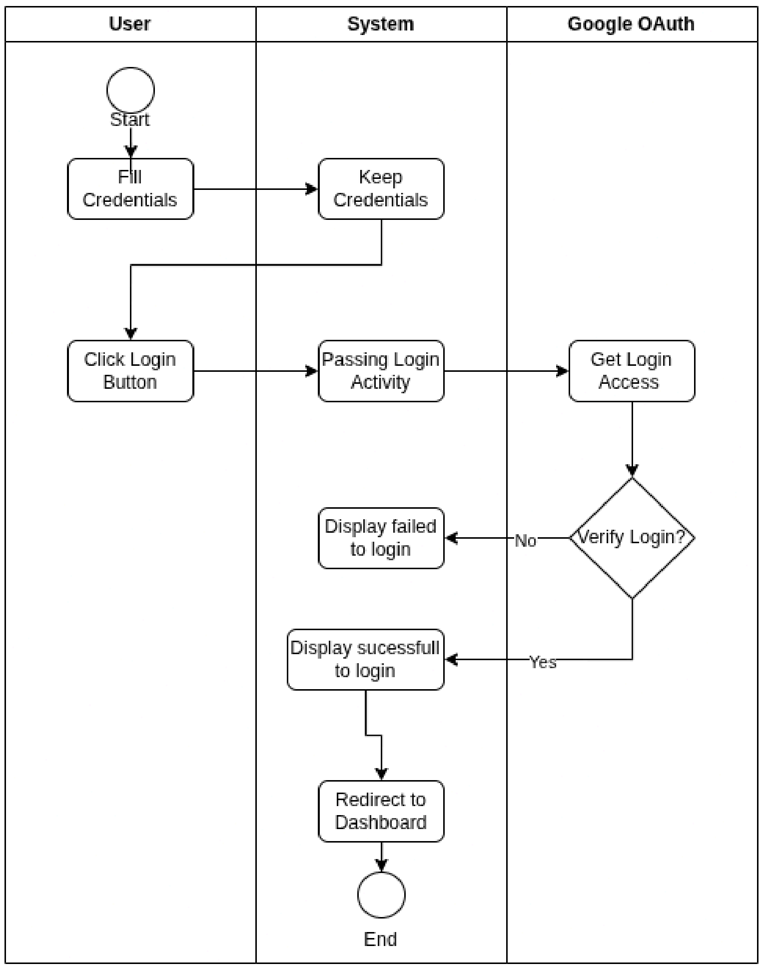
The five modules above are backed by six MySQL tables: users, roles, reports, comments, categories, and locations. This relational structure ensures referential integrity and supports all CRUD operations required by the functional scope.

3.0 Technical Design

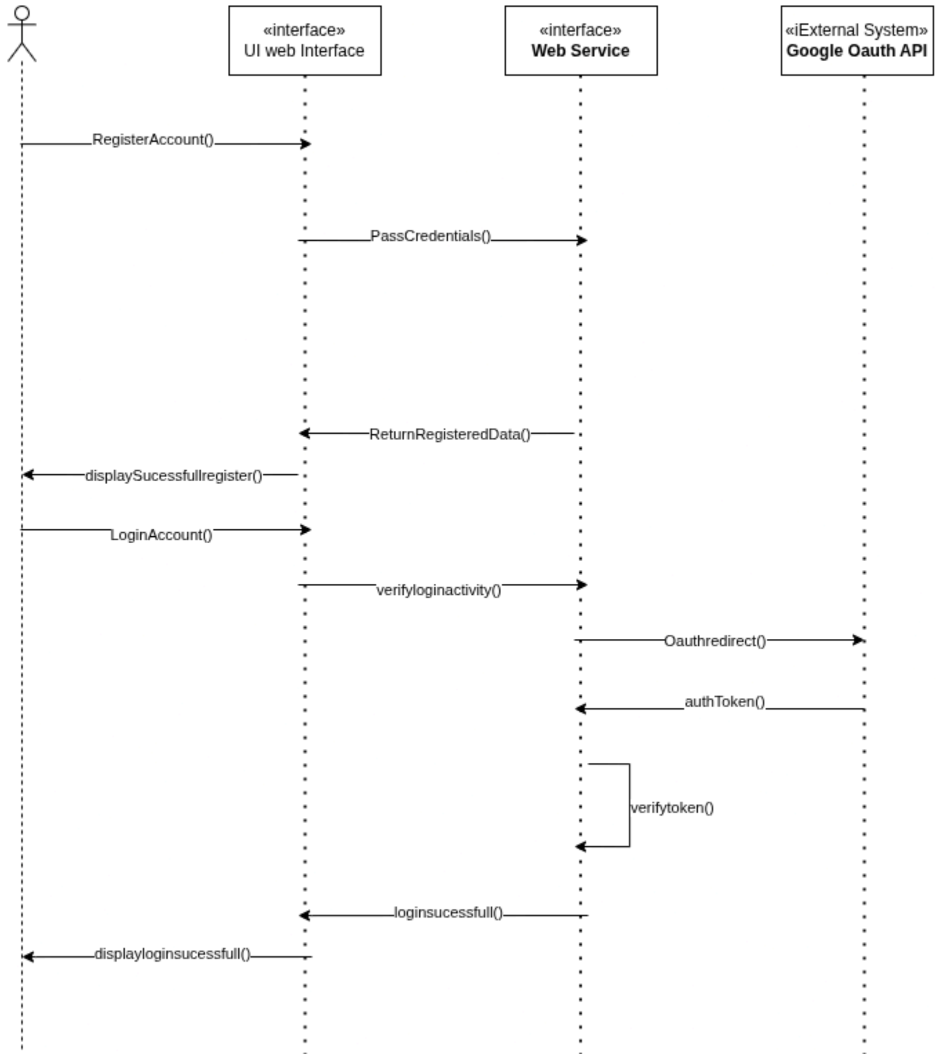
3.1 Entity Relationship Diagram



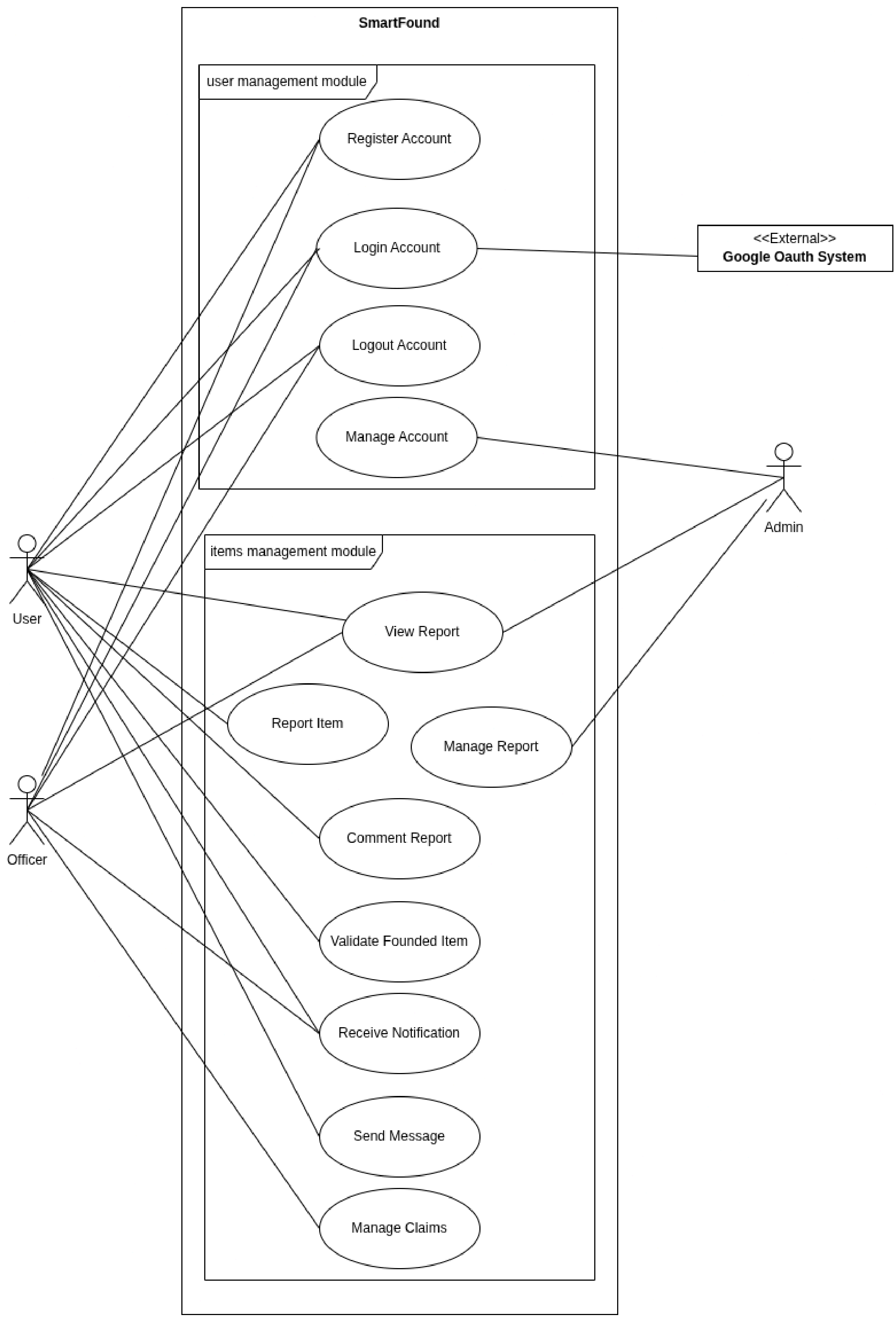
The ERD centers on the User entity, which holds a foreign key to Role for access control, and connects downward to Items — the core transaction table storing item name, category, location, image, and timestamp. Items links to Categories and Locations for classification and place reference, and to Report which captures the report type, date, and status per item. User activity is further tracked through Comment (tied to a report and user), Messages (direct messages from a user), and Notification (referencing item and category for alert targeting), forming a complete data model for the lost-and-found lifecycle from submission to resolution.



The activity diagram models the login flow across three swimlanes: User, System, and Google OAuth. The User begins by filling in their credentials, which the System stores, then the User clicks the Login button to trigger the login activity. The System forwards the request to Google OAuth, which verifies the credentials — on failure it returns a "display failed to login" message back to the System, while on success it signals the System to display a successful login and redirect the User to the Dashboard.



The sequence diagram traces the interaction between the User, UI Web Interface, Web Service, and Google OAuth API across two flows: registration and login. During registration, the User calls RegisterAccount(), the UI passes credentials to the Web Service, which returns the registered data and displays a success message. During login, the UI calls verifyloginactivity() on the Web Service, which redirects to Google OAuth via OAuthredirect(), receives an authToken(), internally calls verifytoken(), and finally returns loginsuccessful() back to the UI, which displays the result to the User.

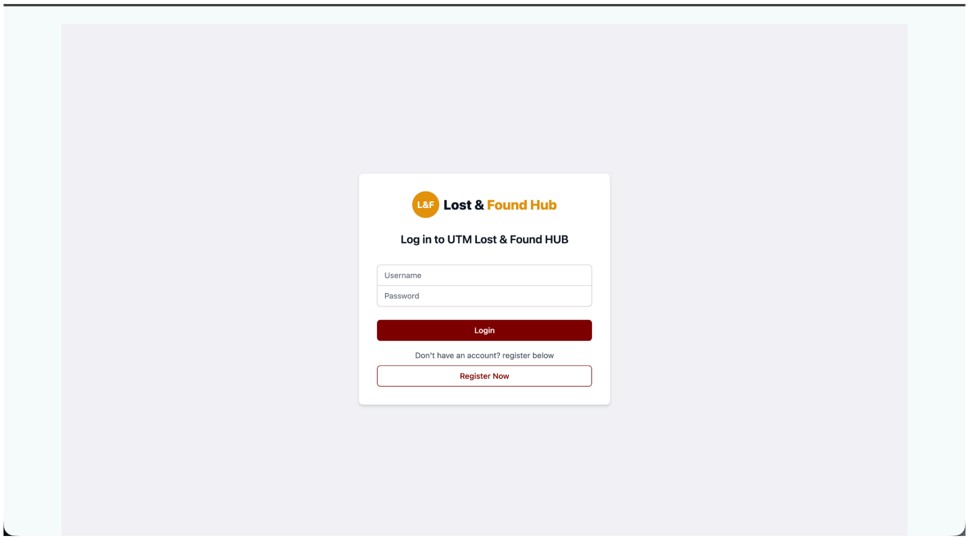

SmartFound
user management module
Register Account
Login Account
Logout Account
Manage Account
<<External>>
Google Oauth System
Admin
User
Officer
items management module
View Report
Report Item
Manage Report
Comment Report
Validate Founded Item
Receive Notification
Send Message
Manage Claims

The use case diagram defines the functional scope of SmartFound across two modules. In the User Management Module, both User and Officer actors can Register Account, Login Account (via Google OAuth System), Logout Account, and Manage Account, while Admin additionally manages accounts. In the Items Management Module, Users can Report Item, View Report, Comment Report, Receive Notification, Send Message, and Manage Claims; Officers share access to most item use cases and can additionally Validate Founded Item; Admin has exclusive access to Manage Report, providing moderation capability over all submitted reports.

4.0 Interface Planning

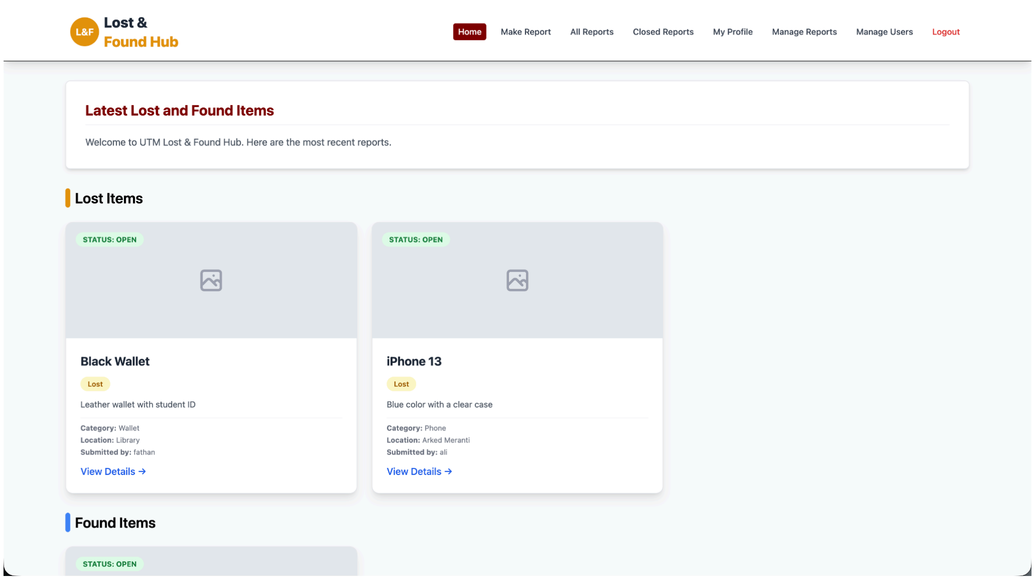
The proposed system architecture features a streamlined navigation flow designed for intuitive user interaction. The user journey commences at the authentication portal (Login or Register), directing authenticated users to a central Home dashboard that highlights the latest lost and found submissions. Core platform functionalities are systematically organized within the main navigation, enabling users to submit new entries through a structured "Make Report" form or explore existing database entries via the "All Reports" and "Closed Reports" directories. Furthermore, the "My Profile" module facilitates personal account management and the tracking of individual submissions, while designated navigation links provide authorized personnel with direct access to advanced administrative management dashboards.

4.1 Login Page



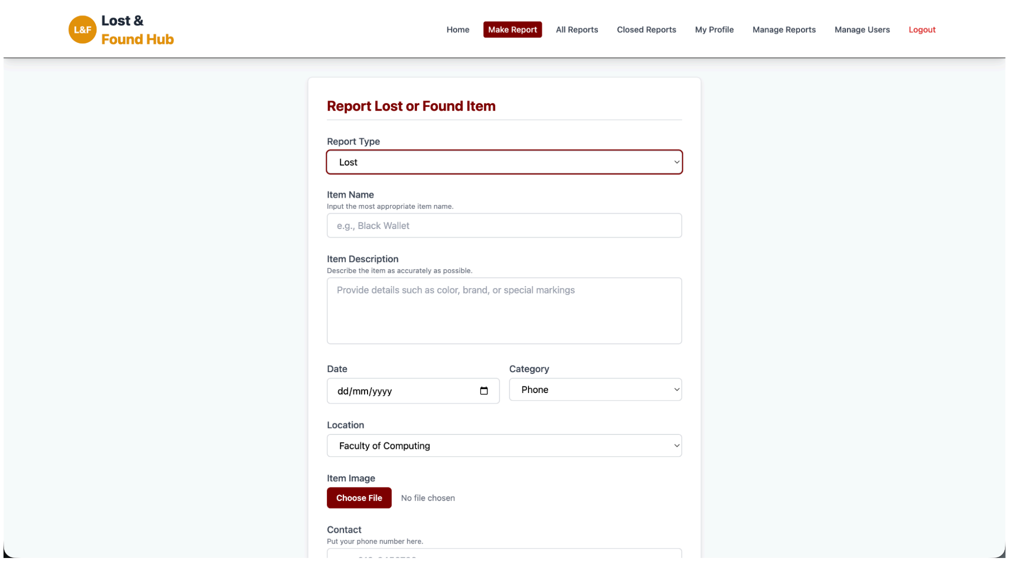
This wireframe illustrates the proposed authentication portal for the UTM Lost & Found Hub. Designed with a clean, centralized layout, the interface provides a straightforward, secure login mechanism requiring standard username and password credentials. To facilitate new user onboarding, the module prominently features a primary "Login" action alongside a clear secondary pathway, enabling unauthenticated visitors to seamlessly transition to the account creation flow via the "Register Now" function.

4.2 Home Page



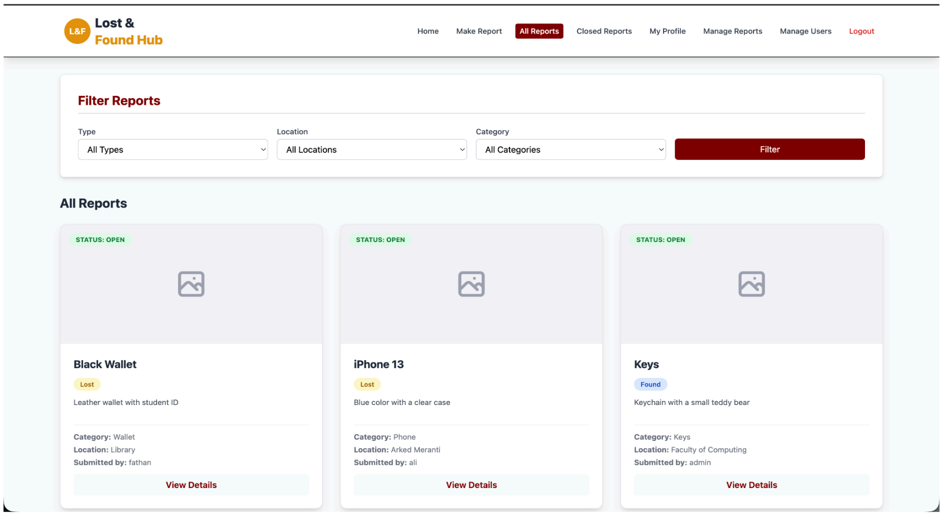
This wireframe illustrates the proposed "Home" landing page for the UTM Lost & Found Hub. Designed to provide an immediate overview of recent platform activity, the dashboard structurally organizes the latest submissions into distinct "Lost Items" and "Found Items" sections.

4.3 Make Report



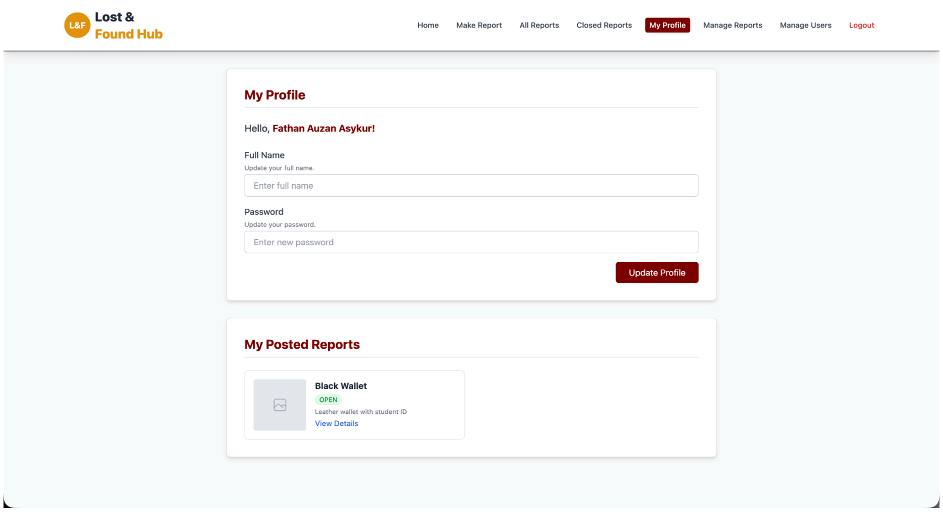
This wireframe illustrates the proposed "Report Lost or Found Item" data entry interface, serving as the primary mechanism for users to submit new entries to the platform. The structured form is designed to capture essential item details systematically, featuring intuitive dropdown menus to categorize the report type (e.g., Lost or Found), item category, and specific location. To ensure comprehensive data collection, it incorporates standardized text inputs for the item name, detailed descriptions, and contact information, alongside a date picker for logging the event timeline and an integrated file upload component for attaching visual evidence of the item.

4.4 All Reports



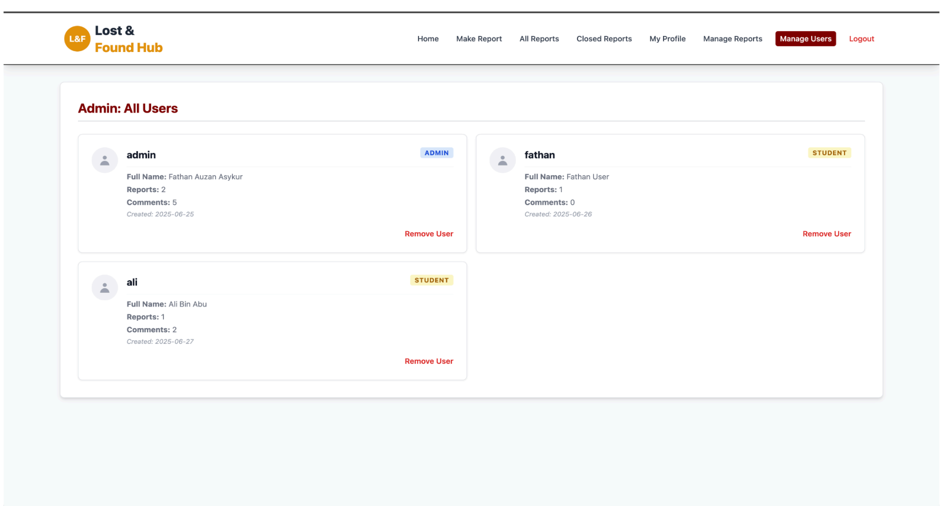
This wireframe depicts the proposed "All Reports" interface for the Lost & Found Hub, serving as the primary discovery dashboard for users. It features a prominent filtering module at the top, allowing users to efficiently refine their searches by item type, location, and category. The main content area utilizes a grid layout to display individual lost and found items as interactive cards. Each card provides a comprehensive overview at a glance, including an image placeholder, current resolution status (e.g., "OPEN"), item classification ("Lost" or "Found"), a brief description, key metadata (category, location, and submitter), and a "View Details" action button to access the full report.

4.5 My Profile Page



This wireframe illustrates the proposed "My Profile" interface for the Lost & Found Hub. It is divided into two primary sections: an account management module that allows users to securely update their personal information, such as their full name and password, and a tracking dashboard titled "My Posted Reports." This dashboard provides a summarized view of the user's personal submissions, displaying key item details, current resolution statuses (e.g., "OPEN"), and quick links to access comprehensive report information

4.6 Admin Dashboard



This wireframe illustrates the proposed "Manage Users" administrative dashboard for the Lost & Found Hub. It utilizes a card-based layout to efficiently display key details for all registered accounts, including credentials, assigned roles, engagement metrics, and account creation dates. To support system moderation, each user profile card features a direct "Remove User" action for streamlined account management.

5.0 Team Planning

5.1 Team Members and Roles

Name	Role	Responsibilities
Tegar Insan Tohaga	Project Manager & QA	Leads the team and coordinates all tasks and deadlines. Manages the Git repository and ensures active contributions from all members. Conducts end-to-end testing of the full application before demo. Prepares and formats the proposal document. Oversees presentation preparation and team communication.
Humaira Sheyla Nurfaiza	Frontend Lead	Sets up the Vue.js Single Page Application including routing, directives, components, and state handling. Develops all main UI pages with responsive HTML and CSS layout. Handles JavaScript DOM manipulation and event handling. Implements jQuery dynamic interaction and asynchronous API calls using Axios or Fetch. Builds all forms with client-side validation and user feedback. Designs wireframes and SPA navigation flow for the proposal.
Fathan Auzan Asykur	Backend Lead	Sets up PHP Slim framework and designs all REST API routes with proper middleware. Builds full CRUD endpoints for items, claims, and admin functions. Implements JWT token generation, validation, and role-based protected route middleware. Writes server-side input validation and HTTP status code handling.
Muhammad Rosyid Ridho Indrianto	Database & Security	Designs the MySQL database schema with at least 4 related tables. Writes all PDO queries supporting CRUD operations. Prepares realistic seed data and exports the schema file. Implements defense against SQL injection, XSS, CSRF, and other common web threats. Deploys the application to a live or test server and writes the README with setup instructions.

5.2 Project Timeline

Week	Date	Tasks	Person In Charge
Week 1	30 Apr — 2 May	Group formation, topic selection, role distribution, Git repo setup	Tegar
Week 1	30 Apr — 2 May	Draft problem statement and user focus	Tegar, Humaira
Week 1	30 Apr — 2 May	Design database schema and API concept	Fathan, Rosyid
Week 2	3 May — 7 May	Complete proposal, architecture diagram, wireframes, tech stack, team planning	All members
Week 2	7 May 2026	PROPOSAL SUBMISSION DEADLINE	Tegar
Week 3	8 May — 14 May	Vue.js SPA setup, routing, login and register pages	Humaira
Week 3	8 May — 14 May	PHP Slim setup, auth routes, JWT implementation	Fathan
Week 3	8 May — 14 May	MySQL schema creation, seed data preparation	Rosyid
Week 4	15 May — 21 May	Build all main UI pages, forms, client-side validation	Humaira
Week 4	15 May — 21 May	Build CRUD endpoints for items and claims	Fathan, Rosyid
Week 4	15 May — 21 May	QA , test frontend forms and navigation	Tegar
Week 5	22 May — 28 May	Axios/Fetch async integration, event handling, jQuery interactions	Humaira
Week 5	22 May — 28 May	Role-based middleware, protected routes, input validation	Fathan
Week 5	22 May — 28 May	PDO queries optimization, security implementation (SQL injection, XSS, CSRF)	Rosyid
Week 5	22 May — 28 May	QA, test SPA navigation and async interactions	Tegar

Week	Date	Tasks	Person In Charge
Week 6	29 May — 4 Jun	Frontend finalization, responsive layout, user feedback, bug fixes	Humaira
Week 6	29 May — 4 Jun	Frontend QA, full frontend testing and fixes	Tegar
Week 6	4 June 2026	FRONTEND PROGRESS EVALUATION	All members
Week 7	5 Jun — 11 Jun	Backend finalization, all endpoints, error handling, HTTP status codes	Fathan
Week 7	5 Jun — 11 Jun	Deployment to live server, README documentation	Rosyid
Week 7	5 Jun — 11 Jun	QA, full API testing using Postman	Tegar
Week 8	12 Jun — 18 Jun	Full system integration testing, frontend to backend to database	Tegar, All
Week 8	12 Jun — 18 Jun	Admin dashboard and officer dashboard finalization	Humaira, Fathan
Week 9	19 Jun — 25 Jun	Final bug fixes, deployment check, live URL verification	Rosyid, Tegar
Week 9	19 Jun — 25 Jun	Presentation slides preparation and demo rehearsal	All members
Week 9	25 June 2026	FINAL DEMO AND PRESENTATION	All members
Week 10	28 June 2026	Peer Evaluation submission	All members (individual)

5.3 Contribution Summary

Name	Primary Contribution Area
Tegar Insan Tohaga	Project coordination, QA testing, proposal, presentation
Humaira Sheyla Nurfaiza	Vue.js SPA, all UI pages, forms, async calls, wireframes
Fathan Auzan Asykur	PHP Slim API, JWT, CRUD endpoints, middleware
Muhammad Rosyid Ridho Indrianto	MySQL schema, PDO queries, security, deployment